



Put It Back in Order

Quick facilitation notes & answer key

Goal

Students fix bugs by re-ordering blocks: first by moving the start block to the top (a program must begin with start), then by swapping two move blocks whose order changes the result at the end of the path. Debugging by re-ordering a sequence is the heart of this subject and a clear C3.2 task, with a small C3.1 rewrite.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: cut Program 1's blocks apart so students can physically slide the start block to the top.

Run it (about 20 min)

1. I DO: point to Program 1. 'The start block is in the middle — Bit doesn't know where to begin. The start block must go first.'
2. WE DO: Exercise 1 — the start block is not first. Exercise 2 — reorder to start at 1, forward 2, forward 1; Bit lands on 4.
3. YOU DO: Exercise 3 — Program 2 lands on 3 because forward 3 hits the end first; swap to back 2, forward 3, and Bit lands on 4.
4. Connect: 'Same blocks, wrong order, wrong result — re-ordering is a kind of fix.'

Answer key (modelled by the run-gate)

Ex 1 — the start block ('start at 1') is in the middle; a program must begin with the start block, so the order is wrong.

Ex 2 — right order: start at 1, forward 2, forward 1. Run: $1 \rightarrow 3 \rightarrow 4$. Bit lands on 4.

Ex 3 — Program 2 buggy order (forward 3, back 2): $3 \rightarrow 5$ (stops at the end) $\rightarrow 3$. Swapped (back 2, forward 3): $3 \rightarrow 1 \rightarrow 4$. Bit lands on 4.

Watch for & differentiate

Common idea in Ex 3: 'the blocks didn't change, so the answer can't change' — the end of the path is what makes the order matter; walk both orders to show it.

Simplify: do Program 1 only (move the start block to the top).

Extend: ask students to find another order of Program 2's blocks and check where Bit lands.

Success indicator: the child puts the start block first in Program 1 (lands on 4) and swaps the moves in Program 2 (lands on 4).